IILM UNIVERSITY

Greater Noida

SCHOOL OF COMPUTER SCIENCE & ENGINEERING

FARMCONNECT

An AI-Enabled Agricultural Marketplace and Farmer Support Platform

B.Tech Final Year Capstone Project Report
Academic Session 2026-27

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Working prototype with documented limitations and pending integrations

Declaration

This report presents the implementation and experimental observations of the FarmConnect capstone project. Measured model results are reported from the project training and testing record; they are not independent certification.

A developed integration is distinguished from successful availability of an external service. Pending components are identified explicitly.

Acknowledgement

We thank Mr. Vikas Kumar Tiwari for his guidance and the School of Computer Science & Engineering, IILM University, Greater Noida, for the opportunity to develop FarmConnect.

We acknowledge the authors of the referenced datasets, research papers, open-source frameworks and agricultural information services.

Abstract

FarmConnect is a role-based web application integrating farmer profiles, marketplace-oriented workflows, agricultural information, government-scheme discovery, mandi price intelligence and AI-assisted crop disease analysis.

The implementation uses HTML, CSS and JavaScript for a single-page frontend, Node.js and Express for authenticated APIs, Firebase Firestore for persistence, and Python with TensorFlow/Keras for image classification.

The implemented multi-crop classifier uses MobileNetV2 transfer learning on a PlantVillage-derived dataset comprising 38 crop/disease classes. The recorded held-out evaluation reports 94.88% accuracy on 8,146 test images, macro precision of 94.13%, macro recall of 92.93%, and macro F1-score of 93.17%.

The mandi module uses the official Government of India data.gov.in API with commodity/location filters, caching and error handling. The verified API returns published records successfully (HTTP 200). Identity-provider checks, payments and several advanced modules remain pending.

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1. Introduction and Problem Definition

Farmers may need separate tools for agricultural guidance, crop disease assessment, local market prices, buyer interactions and account management. FarmConnect aims to provide a unified, understandable platform for these tasks. It supports farmer and buyer roles and separates internal application functionality from data supplied by external services.

1.1 Objectives

• Develop authenticated role-based access and persistent Firestore user profiles.

• Offer a responsive farmer dashboard and clear navigation across modules.

• Classify supported crop-leaf images into healthy/disease categories using a trained deep-learning model.

• Integrate government-published mandi observations with crop, State, district and market filters.

• Provide a scheme discovery foundation with a verified-only publication policy.

• Design future-ready identity, marketplace and payment workflows without falsely claiming completion.

1.2 Project Boundaries

Disease predictions are decision-support results, not laboratory diagnosis. Self-declared profile information does not prove government identity or land ownership. Market-price retrieval depends on external publication and service availability. No successful payment settlement or government identity verification is claimed.

2. Literature Survey and Technical Background

PlantVillage provides labeled plant-leaf imagery frequently used in image-classification research [1]. MobileNetV2 uses inverted residuals and linear bottlenecks, making it an efficient transfer-learning backbone [2]. TensorFlow/Keras supports training and saved-model inference [3]. Official market-price records published through data.gov.in support price exploration [4].
Firebase Firestore provides document-oriented storage [5], while Express supports modular web APIs [6].

Important research considerations include domain shift from controlled leaf photographs to real farms, potential correlation between images of related leaves, class imbalance, uncertainty calibration and upstream public-data outages.

3. Objectives, Scope and Feasibility

3.1 Feasibility

Technical: The selected web and ML technologies are available in the development environment. **Operational:** Independent modules permit staged demonstration. **Economic:** Open-source tools and publicly available data lower prototyping cost; hosting, quotas and authorized verification services would require production budgeting. **Privacy:** The current ordinary profile does not collect full Aadhaar numbers or sensitive identity documents.

3.2 Contribution

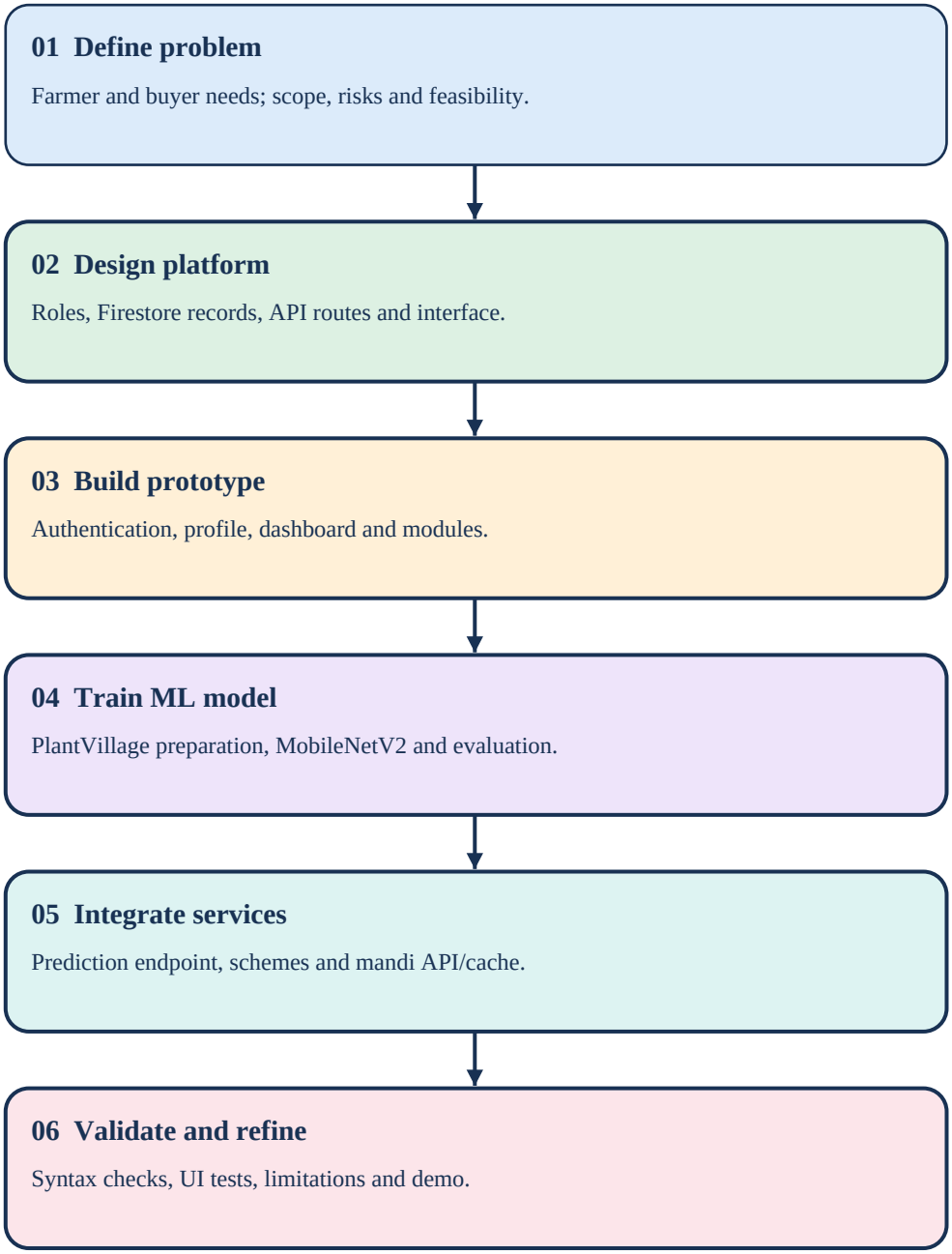
The project contribution is an integrated farmer-oriented software system rather than the invention of a new neural-network architecture. It combines web engineering, cloud persistence, agricultural data integration, user experience and applied machine learning.

4. Requirements and Project Methodology

Area	Implementation / requirement
Frontend	HTML, CSS, JavaScript SPA; farmer/buyer role navigation
Backend	Node.js, Express; REST routes and validation
Database	Firebase Firestore, not MongoDB
ML	Python virtual environment; TensorFlow/Keras; MobileNetV2
Security	Password hashing, token-based authentication and route authorization
Market source	data.gov.in API; published mandi-price observations
Development	VS Code, PowerShell, syntax checks and local testing

4.1 Development Sequence - Project Flowchart

Each phase produces a verifiable output before the next integration step.

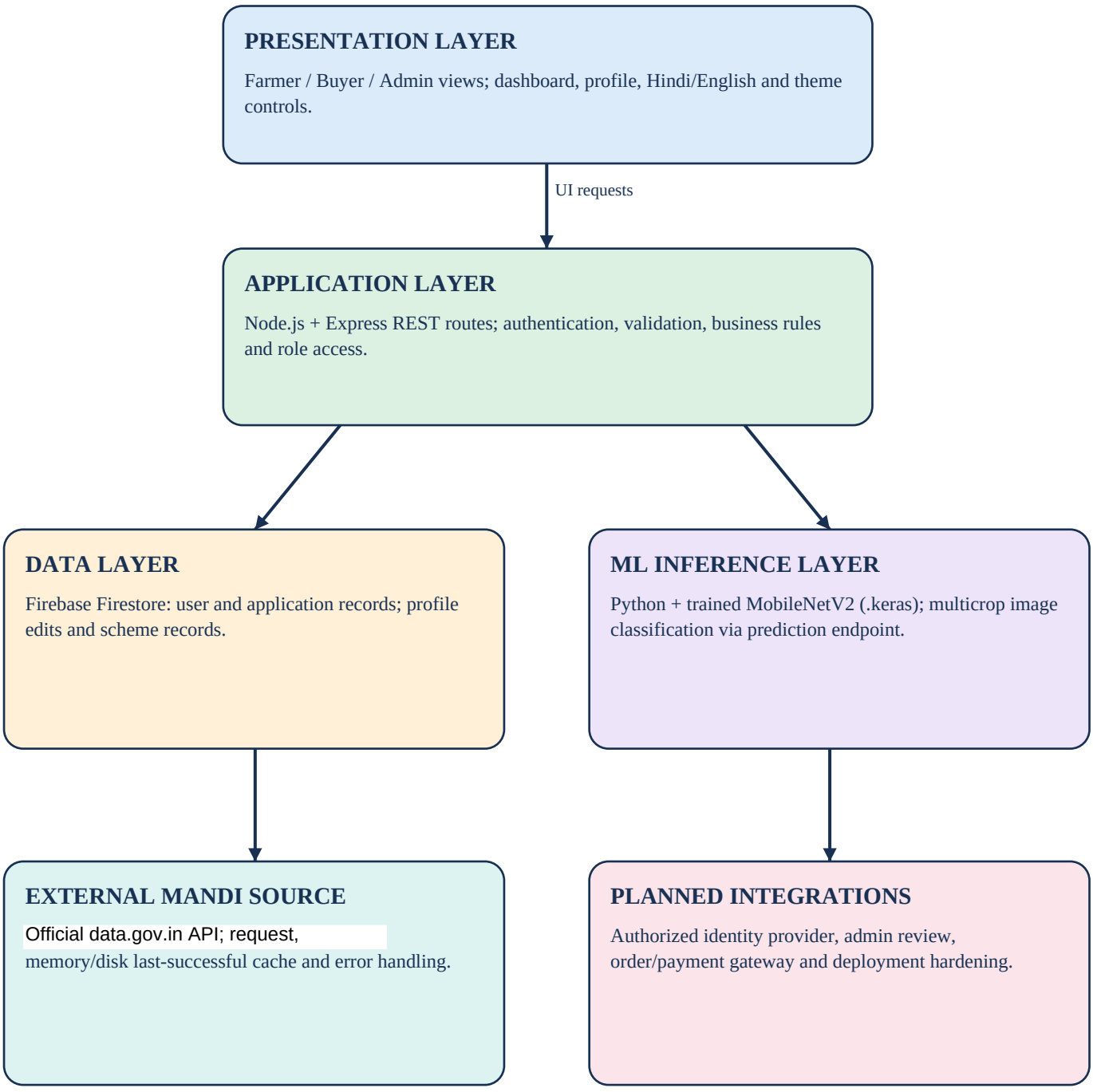


Interpretation

The arrows indicate development dependency, not a claim that all planned production integrations are complete.

5. System Architecture - Layered Design

The diagram shows how the browser, authenticated API, storage and external services interact.



How to read this diagram

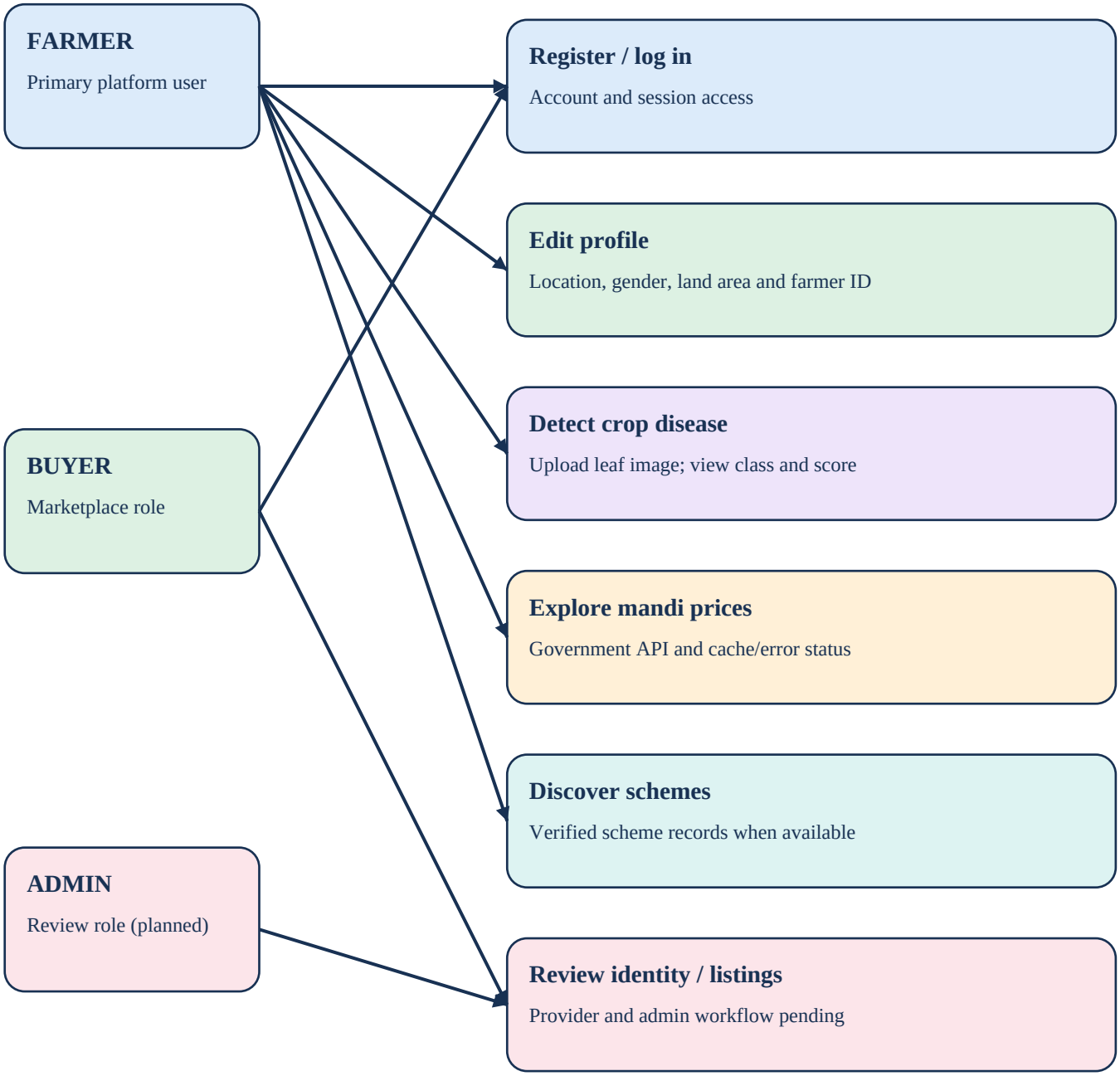
The browser calls protected Express endpoints. The backend reads/writes Firestore and invokes ML or government data services. External failures must not be presented as verified live results.

Implementation boundary

Profile photo is currently stored in the browser only. Government identity verification and payments are planned, not connected.

5.1 Use-Case Diagram - Actors and Responsibilities

Solid-colour cases are implemented/prototyped; rose-colour cases are planned or integration-dependent.

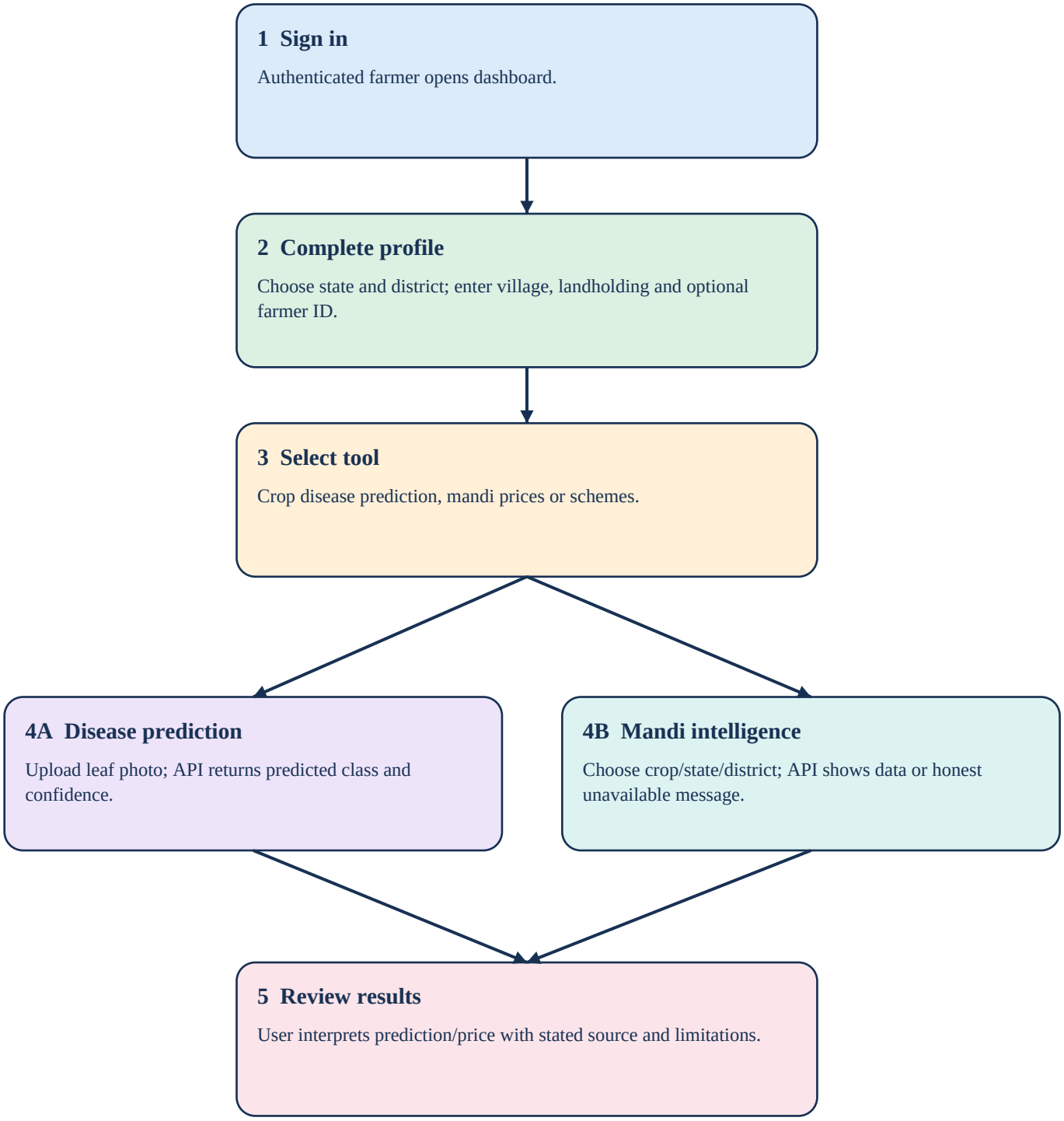


Use-case interpretation

A farmer can maintain a self-declared profile and use agricultural tools. A buyer role exists, while end-to-end marketplace orders, payment records and authorized verification remain future work.

5.2 Farmer Journey - Operational Flowchart

This is the user-facing path from account access to an agricultural decision.



Decision and limitation

ML confidence is a model score, not a laboratory diagnosis. Government mandi data is source-dependent; unavailable upstream responses must not be shown as live prices.

6.3 Scheme Hub and Verification Center

Scheme model/routes and matching infrastructure exist with a verified-only display policy. No verified scheme content is claimed as published. The verification-status endpoint identifies that an authorized identity provider is not connected. A profile or account flag alone is not proof of Aadhaar, landholding or business verification.

7. Machine Learning: Dataset, Models and Methodology

7.1 Dataset and Task

The disease module uses a Kaggle-hosted PlantVillage-derived image collection organized into 38 crop/healthy/disease class folders. The task is multiclass leaf-image classification. The exact Kaggle dataset URL/version and class-to-index mapping should be attached from the training record; the available project notes do not establish these identifiers. A recorded test set contains 8,146 images.

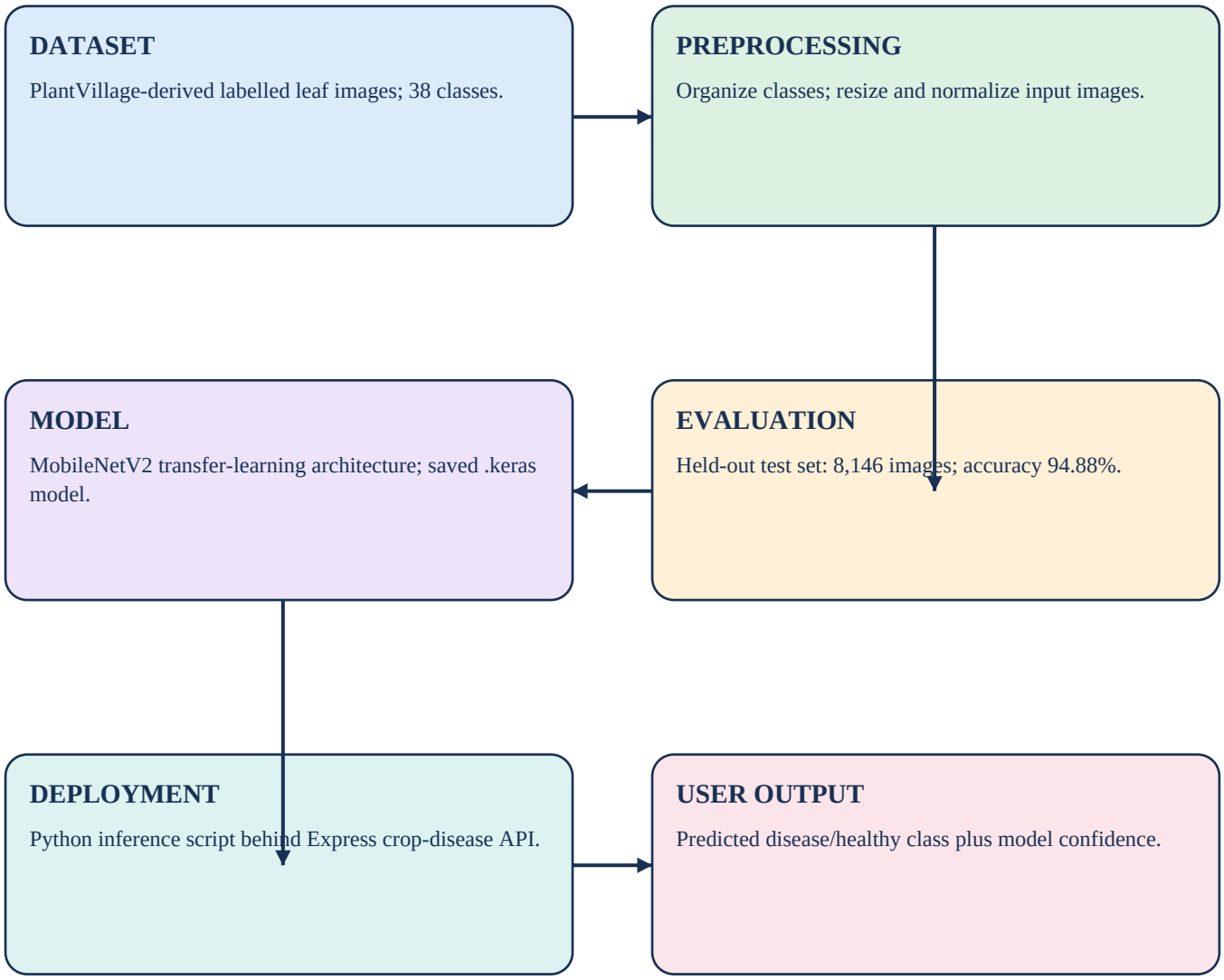
7.2 Models and Libraries Used

Component	Model / method	Purpose
Implemented final disease model	MobileNetV2 transfer learning (CNN)	Feature extraction and 38-class classification
Training / inference framework	TensorFlow and Keras	Training, saved .keras model and prediction
Image processing	Python image preprocessing	Prepare input image for network inference
Inference program	ml/predict_multicrop.py	Load saved model and return predicted class/score
API route	POST /api/crop-disease/predict	Accept image and expose inference to frontend

Model distinction: MobileNetV2 is the documented trained model for the current multi-crop module. Logistic Regression, Naive Bayes, Random Forest and SVM are not asserted as trained or evaluated for FarmConnect. Yield prediction and price forecasting are future modules; no measured accuracy is attributed to them.

7.3 Machine Learning Pipeline - Training and Inference

A labelled multicrop leaf-image dataset supports transfer learning and browser-based prediction.



Recorded test metrics

Accuracy 94.88%; macro precision 94.13%; macro recall 92.93%; macro F1 93.17%. These are recorded project results, not an independently reproduced evaluation.

Confidence and limitations

A displayed confidence such as 98.6% is the model output for an individual image, not overall accuracy. Real-field validation and leakage-aware split checks remain pending.

The 94.88% accuracy is the recorded proportion of correct predictions on the test images, whereas 98.6% is the model score for one Peach bacterial spot example. It is not the overall accuracy, proof of clinical certainty or a guarantee for field photographs. Precision, recall and F1 are reported as macro averages, giving each class equal weight in the aggregate.

9. Mandi Price Intelligence

The mandi module includes crop/commodity, State/UT, district and optional market filtering. It uses the Government of India Open Government Data Platform (data.gov.in), resource ID 35985678-0d79-46b4-9ed6-6f13308a1d24 [4]. Where reported, prices are in rupees per quintal. These are published market observations, not continuous streaming quotes.

9.1 Implemented Integration

• Authenticated Express mandi route and frontend lookup interface.

• Official-source query construction, filtering and published-data labeling.

• In-memory caching and last-successful disk fallback design.

• User-visible error message when official service is temporarily unavailable.

• District selection support and manual fallback where needed.

9.2 External Availability

An earlier resource returned HTTP 500 during development. The corrected official resource has since returned HTTP 200 with published records. The application integration is working; continuous service availability and same-day prices are not guaranteed. The disk fallback displays genuine last-successful data only after a successful retrieval has populated it. Preserve source arrival dates and fetched timestamps, and label cached observations clearly.

9.3 Price Prediction Status

No trained mandi price-forecasting ML model or measured forecasting accuracy is evidenced in the current project record. The current feature is government-data retrieval and filtering, not a predictive-price model.

10. Testing, Validation and Risks

Area	Observed status / next verification
JavaScript syntax	node --check used during frontend/backend patches
Profile fields	Firestore serialization and PATCH allowlist updated; full regression advised
State/district	Local JSON linked; Bihar district options observed in browser
Photo	Local picker and preview; browser-only storage
Crop model	Example prediction and aggregate test metrics recorded
Mandi	Interface/backend implemented; external upstream HTTP 500 encountered
Schemes	Verified-only route policy; verified content pending
Identity / payments	No production integration claimed

10.1 Risks and Mitigation

• External API outage: display honest errors, retry and use only genuine cached observations.

• Dataset shift or leaf-level leakage: test on field images and use stricter split methodology.

• Incomplete administrative location data: retain manual entry and update local dataset.

• Sensitive identity data: require authorized provider and consent before document collection.

• Firestore access: audit server-side authorization and security rules.

• Browser-local photo: label storage scope clearly; cloud sync requires separate consent.

11. Prototype Screenshot and Demonstration

The following original-colour figure reproduces the user-supplied FarmConnect farmer dashboard. It illustrates the navigation, farmer account context, language/theme controls, profile access and mandi-price entry point. A screenshot alone does not establish production readiness of every linked feature.

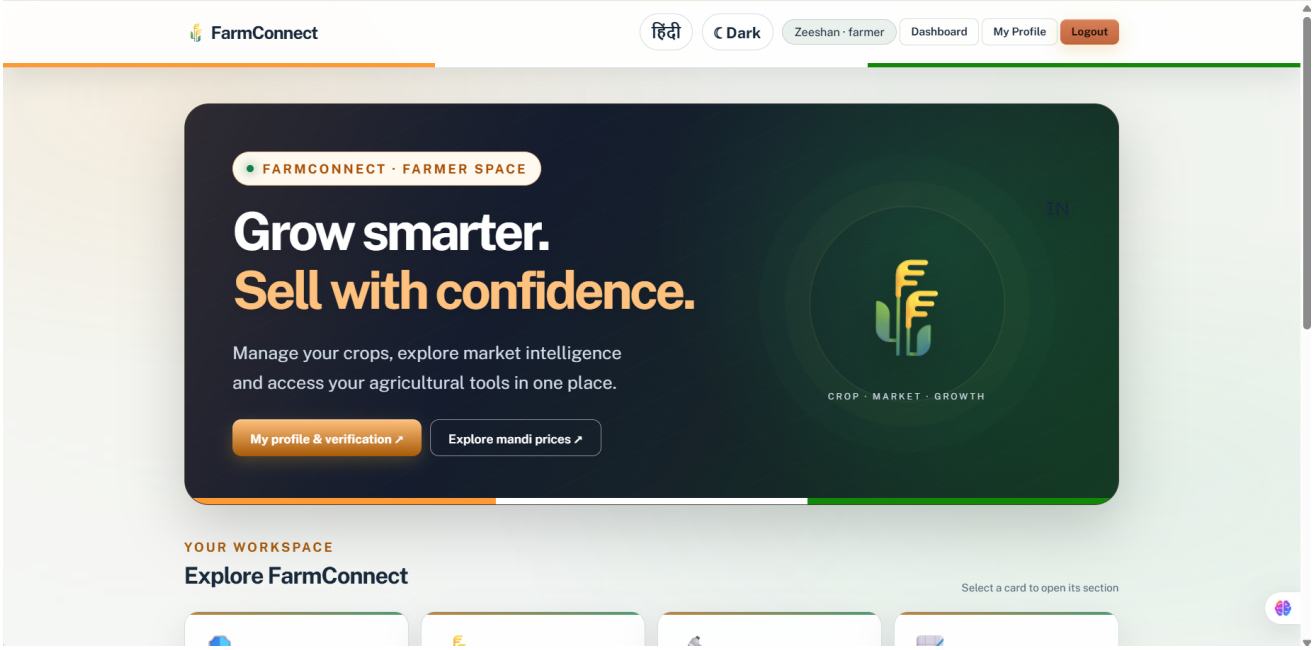


Figure 6. FarmConnect farmer dashboard prototype (original user-supplied colour screenshot).

11.1 Demonstration Sequence

• Log in as farmer and open the dashboard.

• Open Profile; show photo picker, linked State/District, land fields and Firestore save.

• Explain self-declared information versus government verification.

• Upload a supported leaf image; show class and example confidence.

• Open mandi filters; show published data if the upstream responds, otherwise its honest service error.

• Present the architecture, recorded model results, limitations and pending work.

12. Completed and Pending Work

Workstream	Current status	Remaining work
Core dashboard and auth	Implemented prototype	Regression and accessibility review
Firestore farmer profile	Fields and saving route implemented	Security/rule regression
Photo	Browser-local upload	Optional consented cloud sync
State / district	Local JSON dropdown working	Refresh dataset periodically
Village / locality	Manual input	Reliable district-wise suggestions
Crop disease ML	MobileNetV2 trained; inference working	Field-image validation and calibration
Mandi prices	Working official API integration	Monitor API and dated records
Government schemes	Models/routes and verified-only policy	Verified records ingestion
Identity / Aadhaar / land	Provider not connected	Consent and authorized integration
Payments / invoices	Not production-integrated	Gateway, webhooks and ledger
Yield / price forecasting	Planned	Dataset, model and evaluation

12.1 Detailed Remaining Development Roadmap

Priority 1 - Stability and data correctness: Monitor official data.gov.in API responses; preserve genuine dated cache entries; verify district data against current administrative changes; add district-dependent village/locality suggestions with manual fallback; run end-to-end tests for profile persistence, rate display and API failure handling.

Priority 2 - Secure verification and administration: Design explicit consent, authorized identity-provider integration, masked identity references, landholding evidence review, administrator permissions, review audit logs and truthful pending/approved/rejected status. Do not store raw Aadhaar in ordinary profiles or automatically label a self-declaration government verified.

Priority 3 - Marketplace completion: Implement verified produce listings, buyer search, enquiries/bids, order lifecycle, stock updates, dispute handling, payment-gateway sandbox, signed webhook verification, refunds, genuine invoice generation and transaction reconciliation.

Priority 4 - Information modules: Ingest and verify government scheme records; test eligibility matching and expiry dates; add reliable agricultural advisory and notification sources with timestamps and citations.

Priority 5 - ML extensions: Collect independent field-leaf photographs, examine split leakage, evaluate per-class confusion and probability calibration; develop yield analysis only after gathering crop/soil/weather/area/season data and a measurable baseline; develop price forecasting only with a

dated historical market dataset and time-based validation. No future-model accuracy is asserted.

Priority 6 - Production readiness: Security audit, Firestore rules, secrets management, rate limiting, accessibility, multilingual review, monitoring, backups, hosting, load testing, user acceptance and privacy policy.

13. Conclusion and Future Scope

FarmConnect demonstrates an integrated farmer-facing web prototype combining Firestore-backed authenticated profiles, linked State/District selection, an implemented official-data mandi module and a 38-class MobileNetV2 crop disease classifier. The recorded disease-model test accuracy is 94.88% on 8,146 images. The corrected official mandi API has returned records; ongoing availability is not guaranteed. Other pending work includes identity verification, verified schemes, payments and predictive extensions. Priorities are regression and security testing, data sourcing and field-image validation, authorized verification and production transaction handling.

14. References

[1] Hughes, D. P. and Salathe, M., An open access repository of images on plant health to enable the development of mobile disease diagnostics, arXiv:1511.08060 (2015).

[2] Sandler, M. et al., MobileNetV2: Inverted Residuals and Linear Bottlenecks, CVPR (2018).

[3] TensorFlow/Keras documentation, https://www.tensorflow.org/

[4] Government of India, Open Government Data Platform / data.gov.in, resource 35985678-0d79-46b4-9ed6-6f13308a1d24, https://data.gov.in/

[5] Firebase, Cloud Firestore documentation, https://firebase.google.com/docs/firestore

[6] Express documentation, https://expressjs.com/

[7] Kaggle, PlantVillage-derived crop disease image dataset; exact training version/link to be supplied from project notebook.

[8] codingmation, Indian States and Districts JSON, community-maintained GitHub dataset (not official identity verification).

[9] FarmConnect development records and user-supplied dashboard prototype, October 2026.

Appendix: Viva and Submission Notes

• Why Firestore? The existing User model persists documents through Firebase Firestore, not MongoDB.

• Why MobileNetV2? Efficient transfer-learning backbone for 38-class crop disease image classification.

• Accuracy versus confidence: 94.88% is aggregate test accuracy; 98.6% is one prediction score.

- Is mandi live? Official published records load; arrival dates may be historical and API uptime is demonstrated separately.

• Is identity verified? No authorized government identity provider is connected.

• Attach the exact dataset link, training notebook, class mapping, confusion matrix and saved-model version when available.